

A Project report on

ONLINE ASSIGNMENT PLAGIARISM CHECKER

Submitted in partial fulfillment of the requirements

for the award of the degree of

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in

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by

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(B. Tech program accredited by NBA)

SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY: ANANTAPURAMU

(Accredited By NAAC With 'A' Grade, Affiliated To JNTUA, Approved By AICTE, New Delhi)

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Certificate

This is to certify that the project report entitled **ONLINE ASSIGNMENT PLAGIARISM CHECKER** is the bonafide work carried out by **K.PAVANI** bearing Roll Number **164G1A0567**, **G.RAJYALAKSHMI** bearing Roll Number **164G1A0573**, **A.SRUTHI** bearing Roll Number **164G1A05A5**, **S.TARAKESHWAR** bearing Roll Number **164G1A05B1** in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science & Engineering** during the academic year 2019-2020.

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The results embodied in this project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

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LIST OF ABBREVIATIONS

XML	Extensible Markup Language
CLR	Common Language Runtime
IL	Intermediate Language
CTS	Common Type System
CLS	Common Language Specification
OS	Operating System
SQL	Structured Query Language
NLP	Natural Language Processing
MDR	Match Detect Reveal
RAM	Random Access memory
UI	User Interface
DAL	Data Access Layer
BAL	Business Logic Layer
SAAS	Software as a Service
DFD	Data Flow Diagram
UML	Unified Modeling Language

ABSTRACT

Plagiarism is one of the growing issues in academia. Academic Staff faces difficulties in marking student's assignments with higher degree of judgment and waste their valuable time for plagiarism detection. Here we mainly focus on building an effective, simple and fast tool for plagiarism detection on text based assignments to minimize this issue and to help the academic staff in conducting proper evaluation of assignments.

CHAPTER-1

INTRODUCTION

1.1 Motivation

Plagiarism is defined as to take or theft some work and present it has one's own work. Plagiarism is known as illegal use of other's part of work as their own. Plagiarism diminishes one's innovative thinking, creativeness and improvement of knowledge and also it is considered as illegal act in a moral society.

Plagiarism is one of the growing issues in academia. Academic staff faces difficulties in marking student's assignments with higher degree of judgment and waste their valuable time for plagiarism detection. Here we focuses on building an effective, simple and fast tool for plagiarism detection on text based electronic assignments to minimize this issue and to help the academic staff in conducting proper evaluation of assignments.

1.2 Problem Definition

Identifying the Plagiarism between the documents is difficult as well as time consuming so the process should be automated by this it will reduces the time. In the present scenario it has become an enormous problem for any educational institution. It has been a major problem for the faculty for checking plagiarism. Plagiarism checking is done by the faculty by manually which requires much time.

The web application entitled "Online Assignment Plagiarism Checker" scans and Compare the content between documents, displays percentage of Plagiarism detected. Also, students can view the history of their previous documents.

1.3 Objective of Project

The objective of this project is to reduce the time and man power by developing an web application which uses for Plagiarism checking. Also the continuous scanning is an effective method to reduce time and 100% of accuracy.

In this plagiarism checker software, user can register with their basic registration details and create a valid login id and password. By using login id and password, students can login into their personal accounts. After that students can upload assignment file, web application scans and Compare the content between documents, displays percentage of Plagiarism detected.

1.4 Limitations of Project

Limitations of project are:

- It requires an active internet connection.
- The application allows only authenticated users.
- If user uploads an incorrect document, then the result won't be accurate.

CHAPTER-2

LITERATURE SURVEY

2.1 Introduction

In this modern time, with the advancement of internet, easy availability of the computers over the globe has made it easy to access other's work which results in plagiarism. Plagiarism is known as the act of using someone else work without the information of author or without giving acknowledge to that corresponding person. With the advancement of technology, use of computers is growing very vastly and it can be seen that they are used everywhere in schools, institutes and industries. More often, assignments of students are submitted in electronic forms. As e-form is easy and suitable for teachers and students as well, but it leads towards the easy opportunity of plagiarism.

With the widespread of information over the globe, it is very easy to copy the data from different sources which includes internet, papers, books over the internet, newspapers etc. and paste it in a single work without giving any acknowledge to the sources. These actions lead towards lack of learning in students. So there is a need of detecting the plagiarism to increase and improve the learning of students. Plagiarism can occur in any type of field e.g. novels, program's source codes, research papers etc. Furthermore, there can occur in numerous situations when students from different institutes copy the data from internet, different books, journals etc. without quoting any reference. Students, sometimes, do this intentionally but mostly they do it unintentionally due to lack of awareness about the usage of resources according to their own need. The issue is not only confined to the written text but programming codes are also include in it. Different small parts of codes are copied from the source and used according to the requirement without referencing the owners.

2.2 Existing Systems

In existing system, author used Anti-Plagiarism techniques to detect plagiarism only to compare between the database and document.

- ❖ During the survey it has been observed that the existing systems have lot of features that can be extended.
- ❖ Plag Tracker is one of the existing system which will take text as input and compare with web content.
- ❖ Plag Tracker is not 100% accurate, one may need to use an extra plagiarism detection tool to make sure your content is authentic.
- ❖ The present system cannot identify plagiarism from a non-online source.

2.3 Disadvantages of Existing System

- Existing system requires more man power to identify plagiarism.
- Results may not be accurate.
- There is no comparison between the documents of the students.

2.4 Proposed Systems

2.4.1 Online Assignment Plagiarism Checker

The proposed system is “Online Assignment Plagiarism Checker”. It is an web application. This application requires an Active Internet Connection to scan the information. It provides authentication for the students, faculty in the college.

- ❖ We present the application mainly to check the plagiarism between the assignments of the students.
- ❖ It will compare the assignment of each student with other students and display the percentage of plagiarism detected.
- ❖ Our final goal is to make the faculty to identify the students who have copied and based on that they will assign the marks to them.

2.5 Software Development Tool

2.5.1 Introduction to .NET

Microsoft .NET is a set of Microsoft software technologies for rapidly building and integrating XML Web services, Microsoft Windows-based applications, and Web solutions. The .NET Framework is a language-neutral platform for writing programs that can easily and securely interoperate. There's no language barrier with .NET: there are numerous languages available to the developer including Managed C++, C#, Visual Basic and Java Script. The .NET framework provides the foundation for components to interact seamlessly, whether locally or remotely on different platforms. It standardizes common data types and communications protocols so that components created in different languages can easily interoperate.

“.NET” is also the collective name given to various software components built upon the .NET platform. These will be both products (Visual Studio.NET and Windows.NET Server, for instance) and services (like Passport, .NET My Services, and so on).

The .NET Framework

The .NET Framework has two main parts:

1. The Common Language Runtime (CLR).
2. A hierarchical set of class libraries.

The CLR is described as the “execution engine” of .NET. It provides the environment within which programs run. The most important features are:

- ◆ Conversion from a low-level assembler-style language, called Intermediate Language (IL), into code native to the platform being executed on.
- ◆ Memory management, notably including garbage collection.
- ◆ Checking and enforcing security restrictions on the running code.
- ◆ Loading and executing programs, with version control and other such features.

The following features of the .NET framework are also worth description:

Managed Code

The code that targets .NET, and which contains certain extra information - "metadata" - to describe itself. Whilst both managed and unmanaged code can run in the runtime, only managed code contains the information that allows the CLR to guarantee, for instance, safe execution and interoperability.

Information - "metadata" - to describe itself. Whilst both managed and unmanaged code can run in the runtime, only managed code contains the information that allows the CLR to guarantee, for instance, safe execution and interoperability.

Managed Data

With Managed Code comes Managed Data. CLR provides memory allocation and deallocation facilities, and garbage collection. Some .NET languages use Managed Data by default, such as C#, Visual Basic.NET and JScript.NET, whereas others, namely C++, do not. Targeting CLR can, depending on the language you're using, impose certain constraints on the features available. As with managed and unmanaged code, one can have both managed and unmanaged data in .NET applications - data that doesn't get garbage collected but instead is looked after by unmanaged code.

Common Type System

The CLR uses something called the Common Type System (CTS) to strictly enforce type-safety. This ensures that all classes are compatible with each other, by describing types in a common way. CTS define how types work within the runtime, which enables types in one language to interoperate with types in another language, including cross-language exception handling. As well as ensuring that types are only used in appropriate ways, the runtime also ensures that code doesn't attempt to access memory that hasn't been allocated to it.

Common Language Specification

The CLR provides built-in support for language interoperability. To ensure that you can develop managed code that can be fully used by developers using any programming language, a set of language features and rules for using them called the Common Language Specification (CLS) has been defined. Components that follow these rules and expose only CLS features are considered CLS-compliant.

The Class Library

.NET provides a single-rooted hierarchy of classes, containing over 7000 types. The root of the namespace is called System; this contains basic types like Byte, Double, Boolean, and String, as well as Object. All objects derive from System. Object. As well as objects, there are value types. Value types can be allocated on the stack, which can provide useful flexibility. There are also efficient means of converting value types to object types if and when necessary.

The set of classes is pretty comprehensive, providing collections, file, screen, and network I/O, threading, and so on, as well as XML and database connectivity.

The class library is subdivided into a number of sets (or namespaces), each providing distinct areas of functionality, with dependencies between the namespaces kept to a minimum.

Languages Supported By .NET

The multi-language capability of the .NET Framework and Visual Studio .NET enables developers to use their existing programming skills to build all types of applications and XML Web services. The .NET framework supports new versions of Microsoft's old favorites Visual Basic and C++ (as VB.NET and Managed C++), but there are also a number of new additions to the family.

Visual Basic .NET has been updated to include many new and improved language features that make it a powerful object-oriented programming language. These features include inheritance, interfaces, and overloading, among others. Visual Basic also now

supports structured exception handling, custom attributes and also supports multi-threading.

Visual Basic .NET is also CLS compliant, which means that any CLS-compliant language can use the classes, objects, and components you create in Visual Basic .NET.

Managed Extensions for C++ and attributed programming are just some of the enhancements made to the C++ language. Managed Extensions simplify the task of migrating existing C++ applications to the new .NET Framework. C# is Microsoft's new language. It's a C-style language that is essentially "C++ for Rapid Application Development". Unlike other languages, its specification is just the grammar of the language. It has no standard library of its own, and instead has been designed with the intention of using the .NET libraries as its own.

Microsoft Visual J# .NET provides the easiest transition for Java-language developers into the world of XML Web Services and dramatically improves the interoperability of Java-language programs with existing software written in a variety of other programming languages.

Active State has created Visual Perl and Visual Python, which enable .NET-aware applications to be built in either Perl or Python. Both products can be integrated into the Visual Studio .NET environment. Visual Perl includes support for Active State's Perl Dev Kit.

Other languages for which .NET compilers are available include:

- FORTRAN
- COBOL
- Eiffel

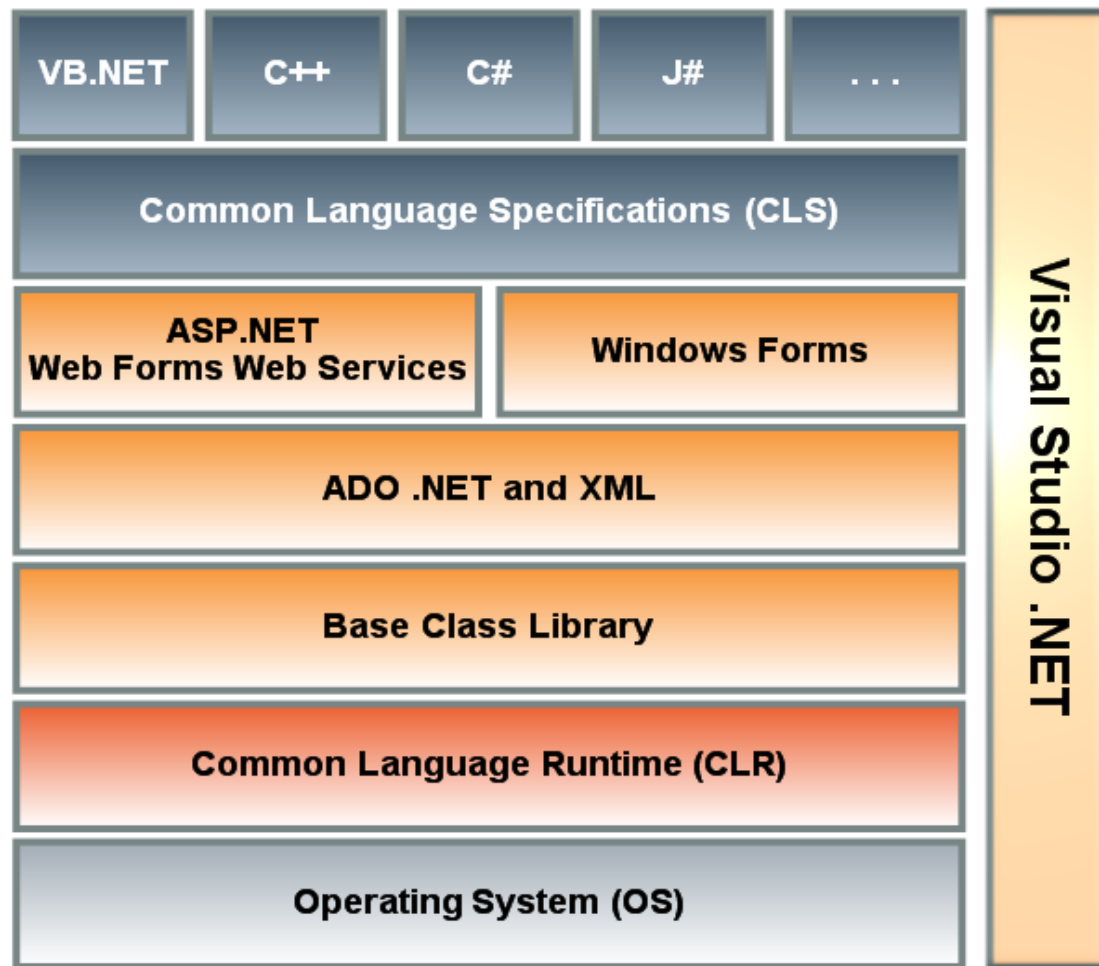


Figure 2.1: .NET Framework

C#.NET is also compliant with CLS (Common Language Specification) and supports structured exception handling. CLS is set of rules and constructs that are supported by the CLR (Common Language Runtime). CLR is the runtime environment provided by the .NET Framework; it manages the execution of the code and also makes the development process easier by providing services.

C#.NET is a CLS-compliant language. Any objects, classes, or components that created in C#.NET can be used in any other CLS-compliant language. In addition, we can use objects, classes, and components created in other CLS-compliant languages in C#.NET .The use of CLS ensures complete interoperability among applications, regardless of the languages used to create the application.

Constructors and Destructors

Constructors are used to initialize objects, whereas destructors are used to destroy them. In other words, destructors are used to release the resources allocated to the object. In C#.NET the sub finalize procedure is available. The sub finalize procedure is used to complete the tasks that must be performed when an object is destroyed. The sub finalize procedure is called automatically when an object is destroyed. In addition, the sub finalize procedure can be called only from the class it belongs to or from derived classes.

Garbage Collection

Garbage Collection is another new feature in C#.NET. The .NET Framework monitors allocated resources, such as objects and variables. In addition, the .NET Framework automatically releases memory for reuse by destroying objects that are no longer in use.

In C#.NET, the garbage collector checks for the objects that are not currently in use by applications. When the garbage collector comes across an object that is marked for garbage collection, it releases the memory occupied by the object.

Overloading

Overloading is another feature in C#. Overloading enables us to define multiple procedures with the same name, where each procedure has a different set of arguments. Besides using overloading for procedures, we can use it for constructors and properties in a class.

Multithreading

C#.NET also supports multithreading. An application that supports multithreading can handle multiple tasks simultaneously, we can use multithreading to decrease the time taken by an application to respond to user interaction.

Structured Exception Handling

C#.NET supports structured handling, which enables us to detect and remove errors at runtime. In C#.NET, we need to use Try...Catch...Finally statements to create exception

handlers. Using Try...Catch...Finally statements, we can create robust and effective exception handlers to improve the performance of our application.

The .NET Framework

The .NET Framework is a new computing platform that simplifies application development in the highly distributed environment of the Internet.

Objectives of .NET Framework

1. To provide a consistent object-oriented programming environment whether object codes is stored and executed locally on Internet-distributed, or executed remotely.
2. To provide a code-execution environment to minimizes software deployment and guarantees safe execution of code.
3. Eliminates the performance problems. There are different types of application, such as Windows-based applications and Web-based applications.

2.5.2 Features of SQL-Server

The OLAP Services feature available in SQL Server version 7.0 is now called SQL Server 2000 Analysis Services. The term OLAP Services has been replaced with the term Analysis Services. Analysis Services also includes a new data mining component. The Repository component available in SQL Server version 7.0 is now called Microsoft SQL Server 2000 Meta Data Services. References to the component now use the term Meta Data Services. The term repository is used only in reference to the repository engine within Meta Data Services SQL-SERVER database consist of six type of objects, they are:

1. TABLE
2. QUERY
3. FORM
4. REPORT
5. MACRO

Table:

A database is a collection of data about a specific topic.

Views of Table:

We can work with a table in two types,

1. Design View
2. Datasheet View

Design View

To build or modify the structure of a table we work in the table design view. We can specify what kind of data will be hold.

Datasheet View

To add, edit or analyses the data itself we work in tables datasheet view mode.

Query:

A query is a question that has to be asked the data. Access gathers data that answers the question from one or more table. The data that make up the answer is either dynaset (if you edit it) or a snapshot (it cannot be edited).Each time we run query, we get latest information in the dynaset. Access either displays the dynaset or snapshot for us to view or perform an action on it, such as deleting or updating.

2.6 Summary

Literature survey helps us to know the functionality about the existing system and provides to create a mobile application on Online Assignment Plagiarism Checker with new features that do not exist already.

CHAPTER-3

ANALYSIS

3.1 Introduction

Plagiarism is defined as the wrongful appropriation or stealing of some other people ideas and make it as own. Stealing or copying of data now a day is becoming very common. Plagiarism detection of copied data originated in 1970's and common methods of Natural language processing (NLP) for detection of copied data introduced in three different techniques namely Grammar-based method, Semantic-based method and Grammar semantic hybrid method. In grammar-based method grammatical structure of the document is maintained and it used a string matching technique for calculating similarity between documents. Semantic based method uses vector space model of the information retrieval technique, and statistical words frequency in document to obtain the vector of the documents, then uses dot product, cousins or other methods to calculate the vectors of two documents.

This featured vector is the similarity of the document. This technique isn't effective as it doesn't give the source of the plagiarized data. Grammar semantic hybrid method it improves the detection result of these two methods. It is important and effective to highlight or mark the plagiarized text in the documents in parallel to the similarity results. In this we proposed the Longest Common Consecutive Word algorithm, it considers the whole paragraph as a single unit and tracks the words positions. Then by-word comparison is carried out and common words are obtained, this gives the plagiarized version and similarity between documents.

MDR (Match Detect Reveal) is the method in which the document whose plagiarism is going to be checked is first split into the fixed length strings by maintaining a suffix tree. String matching algorithm is used for comparison, and longest common strings can be found in suffix tree. By this, the similarity index and location in the documents can be obtained. This technique is not efficient because it uses the exact words that match and hence making the unclear plagiarized text version.

3.2 Software Requirement Specifications

3.2.1 User Requirements

The software requirements specification is produced at the culmination of the analysis task. The function and performance allocated to the software as a part of system engineering and refined by establishing a complex information description, detailed functional and behavioral description, and indication of performance requirements and design constraints, appropriate validation criteria and other data pertinent to requirements.

The major requirement included in this application is to detect the duplicate the data. This is the highest level abstraction of the requirement, this to be converted into lower level requirement language in detail. The programmer has to correctly has understood.

3.2.2 Software Requirements

- Operating system : Windows
- Coding Language : ASP.net, C#.net
- Tool : Visual Studio
- Database : MY SQL

3.2.3 Hardware Requirements

- System : Pentium IV 2.4 GHz.
- Hard Disk : 40 GB.
- Monitor : 1
- Mouse : 1
- RAM : 1

3.3 Visual Studio Installation Procedure

Step 1 - Make sure your computer is ready for Visual Studio

Before you begin installing Visual Studio:

1. Check the [system requirements](#). These requirements help you know whether your computer supports Visual Studio 2019.
2. Apply the latest Windows updates. These updates ensure that your computer has both the latest security updates and the required system components for Visual Studio.
3. Reboot. The reboot ensures that any pending installs or updates don't hinder the Visual Studio install.
4. Free up space. Remove unneeded files and applications from your System Drive by, for example, running the Disk Cleanup app.

For questions about running previous versions of Visual Studio side by side with Visual Studio 2019, see the [Visual Studio 2019 Platform Targeting and Compatibility](#) page.

Step 2 - Download Visual Studio

Next, download the Visual Studio bootstrapper file.

To do so, choose the following button, choose the edition of Visual Studio that you want, choose **Save**, and then choose **Open folder**.

[Download Visual Studio](#)

Step 3 - Install the Visual Studio installer

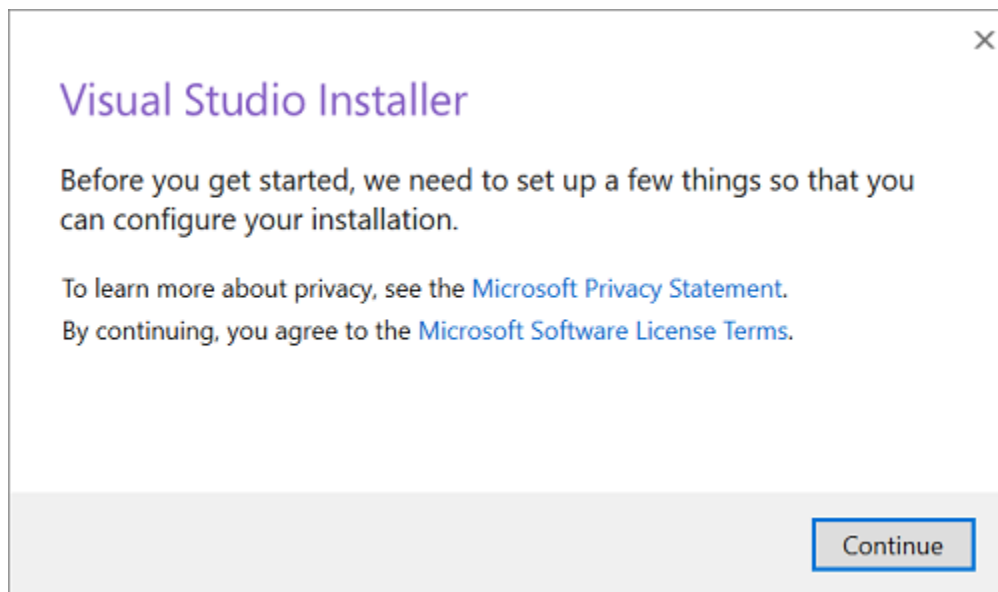
Run the bootstrapper file to install the Visual Studio Installer. This new lightweight installer includes everything you need to both install and customize Visual Studio.

1. From your **Downloads** folder, double-click the bootstrapper that matches or is similar to one of the following files:

- **vs_community.exe** for Visual Studio Community
- **vs_professional.exe** for Visual Studio Professional
- **vs_enterprise.exe** for Visual Studio Enterprise

If you receive a User Account Control notice, choose **Yes**.

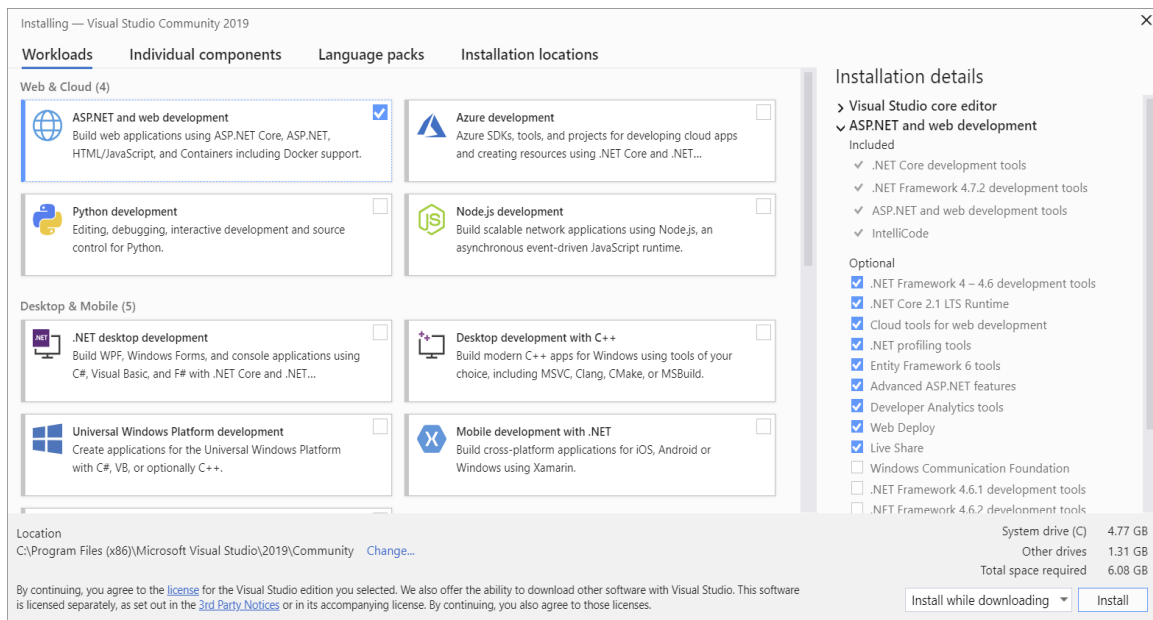
2. We'll ask you to acknowledge the Microsoft [License Terms](#) and the Microsoft [Privacy Statement](#). Choose **Continue**.



Step 4 - Choose workloads

After the installer is installed, you can use it to customize your installation by selecting the feature sets—or workloads—that you want. Here's how.

1. Find the workload you want in the **Visual Studio Installer**.



For example, choose the "ASP.NET and web development" workload. It comes with the default core editor, which includes basic code editing support for over 20 languages, the ability to open and edit code from any folder without requiring a project, and integrated source code control.

2. After you choose the workload(s) you want, choose **Install**.

Next, status screens appear that show the progress of your Visual Studio installation.

At any time after installation, you can install workloads or components that you didn't install initially. If you have Visual Studio open, go to **Tools > Get Tools and Features...** which opens the Visual Studio Installer. Or, open **Visual Studio Installer** from the Start menu. From there, you can choose the workloads or components that you wish to install. Then, choose **Modify**.

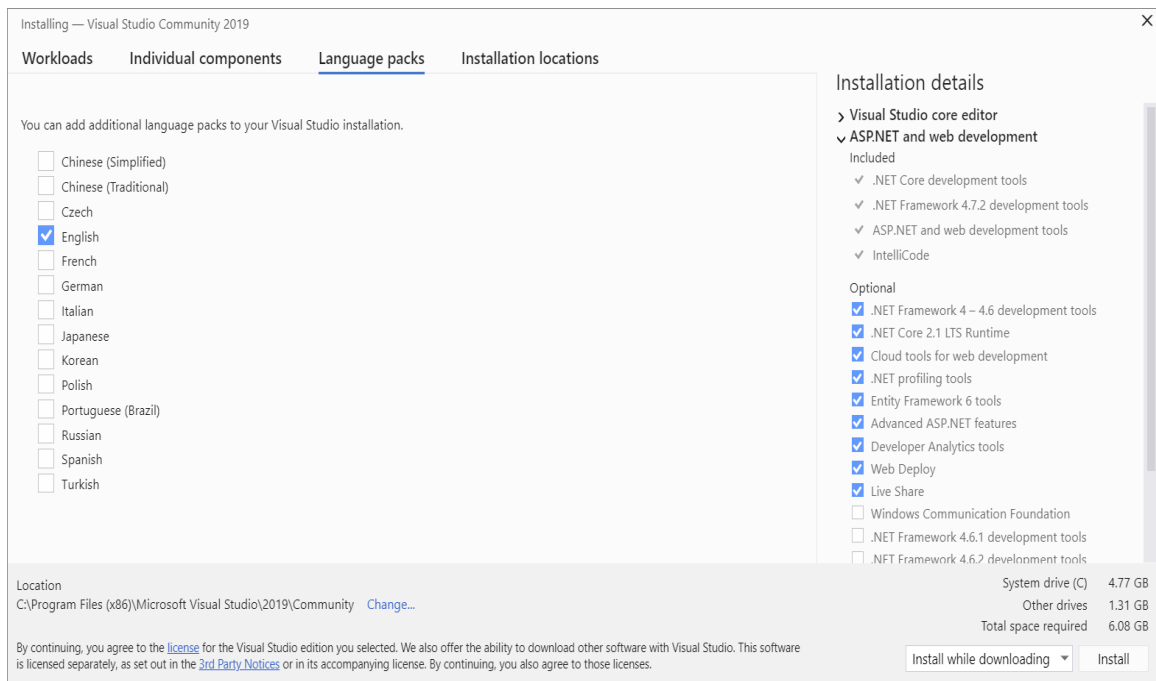
Step 5 - Choose individual components (Optional)

If you don't want to use the Workloads feature to customize your Visual Studio installation, or you want to add more components than a workload installs, you can do so

by installing or adding individual components from the **Individual components** tab. Choose what you want, and then follow the prompts.

Step-6 - Install language packs (Optional)

By default, the installer program tries to match the language of the operating system when it runs for the first time. To install Visual Studio in a language of your choosing, choose the **Language packs** tab from the Visual Studio Installer, and then follow the prompts.

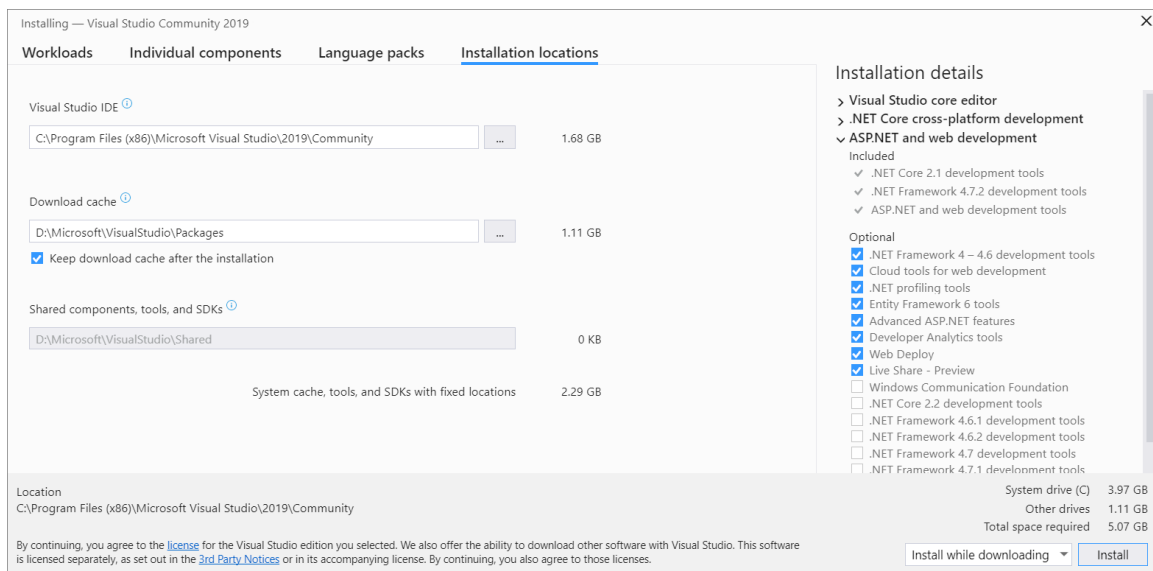


Change the installer language from the command line

Another way that you can change the default language is by running the installer from the command line. For example, you can force the installer to run in English by using the following command: `vs_installer.exe --locale en-US`. The installer will remember this setting when it is run the next time. The installer supports the following language tokens: zh-cn, zh-tw, cs-cz, en-us, es-es, fr-fr, de-de, it-it, ja-jp, ko-kr, pl-pl, pt-br, ru-ru, and tr-tr.

Step 7 - Select the installation location (Optional)

You can reduce the installation footprint of Visual Studio on your system drive. You can choose to move the download cache, shared components, SDKs, and tools to different drives, and keep Visual Studio on the drive that runs it the fastest.



You can select a different drive only when you first install Visual Studio. If you've already installed it and want to change drives, you must uninstall Visual Studio and then reinstall it.

Step 8 - Start developing

1. After Visual Studio installation is complete, choose the **Launch** button to get started developing with Visual Studio.
2. On the start window, choose **Create a new project**.
3. In the search box, enter the type of app you want to create to see a list of available templates. The list of templates depends on the workload(s) that you chose during installation. To see different templates, choose different workloads.

You can also filter your search for a specific programming language by using the **Language** drop-down list. You can filter by using the **Platform** list and the **Project type** list, too.

4. Visual Studio opens your new project, and you're ready to code!

3.4 System study

Feasibility Study

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

1. Economical Feasibility
2. Technical Feasibility
3. Social Feasibility

Economical Feasibility

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

Technical Feasibility

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed

system must have a modest requirement, as only minimal or null changes are required for implementing this system.

Social Feasibility

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

3.5 Summary

The “Online Assignment Plagiarism Checker” application whose specifications have been provided Simple but it is very helpful for the Education institutions. We also see how requirements gathering and design are very critical in software development to make sure that the project solves the right problem in right approach.

CHAPTER-4

DESIGN

4.1 Introduction

System design is the solution to the creation of a new system. This phase is composed of several systems. This phase focuses on the detailed implementation of the feasible system. It emphasis on translating design specifications to performance specification is system design. System design has two phases of development logical and physical design.

During logical design phase the analyst describes inputs (sources), outputs (destinations), databases (data stores) and procedures (data flows) all in a format that meats the uses requirements. The analyst also specifies the user needs and at a level that virtually determines the information flow into and out of the system and the data resources. Here the logical design is done through data flow diagrams and database design.

The physical design is followed by physical design or coding. Physical design produces the working system by defining the design specifications, which tell the programmers exactly what the candidate system must do.

The programmers write the necessary programs that accept input from the user, perform necessary processing on accepted data through call and produce the required report on a hard copy or display it on the screen.

4.2 System Architecture

4.2.1 Architectural Design

3-Tier architecture is also called layered architecture. Some people called it n-tier architecture. Layer architectures are essentially objects and work in object oriented environment. 3-tier architecture is a very well-known architecture in the world of

software development, it doesn't matter whether you are developing web based application or desktop based, it is the best architecture to use.

3-Tier architecture consists of

- 1) UI or Presentation Layer
- 2) Logic Layer
- 3) Data Access Layer

Presentation Layer

Presentation layer consists of pages like .net or desktop based form where data is presented to users or getting input from users.

Logic layer

Logic layer contains all of the Plagiarism logic. Its responsibility is to validate the Plagiarism rules of the component and communicating with the Data Access Layer. Logic Layer is the class in which we write functions that get data from Presentation Layer and send that data to database through Data Access Layer.

Data Access Layer

Data Access Layer is also the class that contains methods to enable business logic layer to connect the data and perform desired actions. These desired actions can be selecting, inserting, updating and deleting the data. DAL accepts the data from BAL and sends it to the database or DAL gets the data from the database and sends it to the business layer. In short, its responsibility is to communicate with the backend structure.

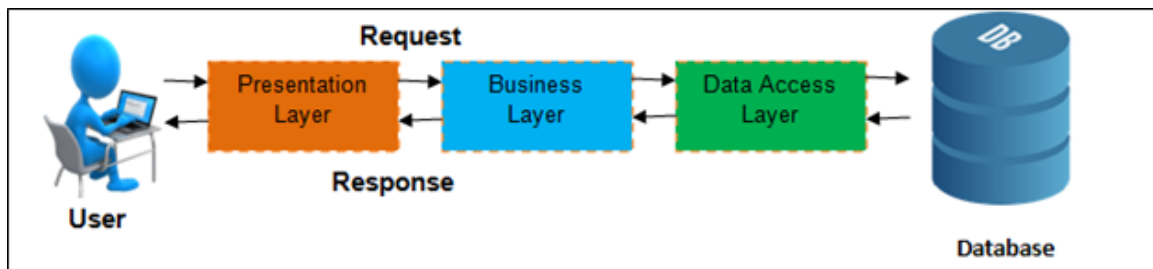


Figure 4.1:3-Tier Architecture

4.2.2 Single Tenant Architecture

Single-tenancy is an architecture in which a single instance of a software application and supporting infrastructure serves one customer. Single-tenancy is commonly implemented in software-as-a-service (SaaS) delivery models or in cloud services. In single-tenancy architectures, a customer -- called a tenant -- will have a singular instance of a SaaS application dedicated to them.

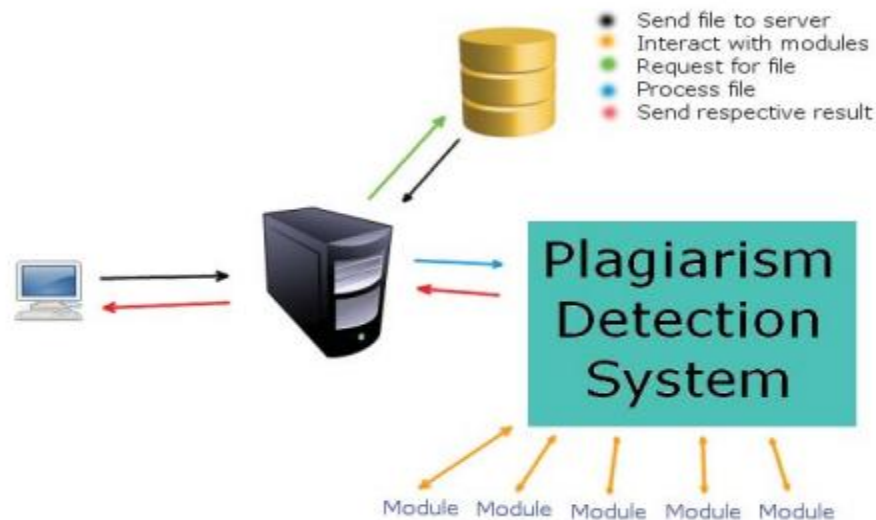


Figure 4.2: Single Tenant Architecture

4.2.3 Multi-Tenant Architecture

Multi-tenant architecture, commonly referred to as Multi tenancy, is a software architecture in which multiple single instances of software run on a single physical server. The server then serves multiple tenants.



Figure 4.3: Multi-Tenant Architecture

4.3 Data Flow Diagram

A data-flow diagram is a way of representing a flow of a data of a process or a system. The DFD also provides information about the outputs and inputs of each entity and the process itself. A data-flow diagram has no control flow, there are no decision rules and no loops.

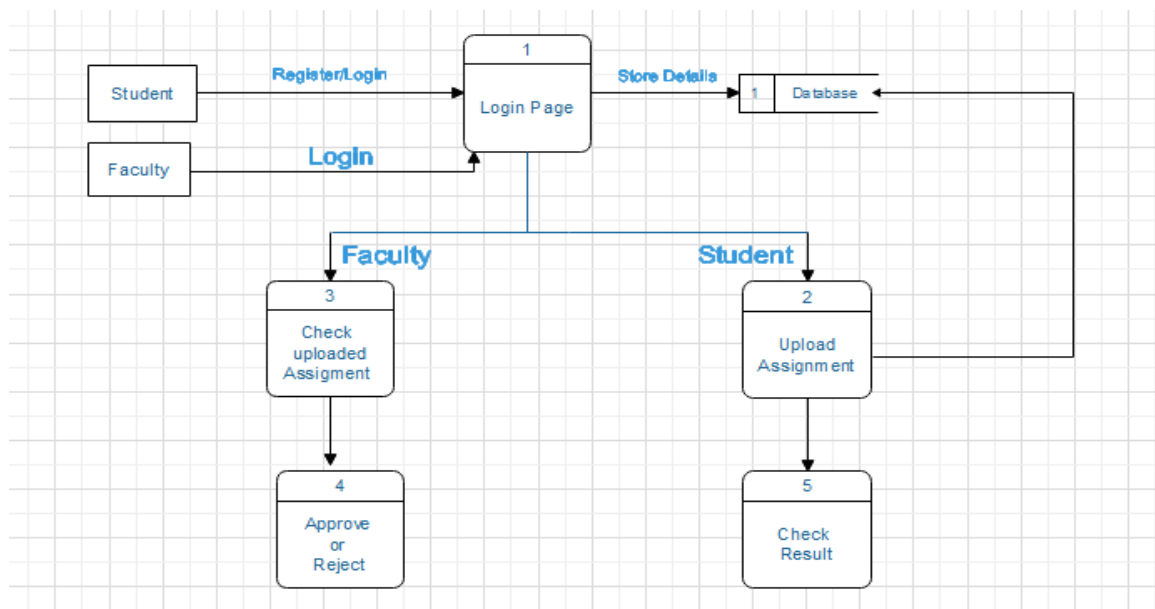


Figure 4.4: Data flow Diagram

4.4 Input Design

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data in to a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining the privacy. Input Design considered the following things:

- What data should be given as input?
- How the data should be arranged or coded?
- The dialog to guide the operating personnel in providing input.
- Methods for preparing input validations and steps to follow when error occur.

Objectives

1. Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system.
2. It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulates can be performed. It also provides record viewing facilities.
3. When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needed so that the user will not be in maize of instant. Thus the objective of input design is to create an input layout that is easy to follow.

4.5 Output Design

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output. It is the most important and direct source information to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making.

1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can use easily and effectively. When analysis design computer output, they should Identify the specific output that is needed to meet the requirements.
2. Select methods for presenting information.
3. Create document, report, or other formats that contain information produced by the system.

The output form of an information system should accomplish one or more of the following objectives.

- ❖ Convey information about past activities, current status or projections of the
- ❖ Future.
- ❖ Signal important events, opportunities, problems, or warnings.
- ❖ Trigger an action.
- ❖ Confirm an action.

4.6 UML Diagrams

4.6.1 Class Diagram

In software engineering, a **class diagram** in the Unified Modeling Language (UML) is a type of static structure **diagram** that describes the structure of a system by

showing the system's **classes**, their attributes, operations (or methods), and the relationships among objects. Class Diagram is shown in Figure no.4.5.

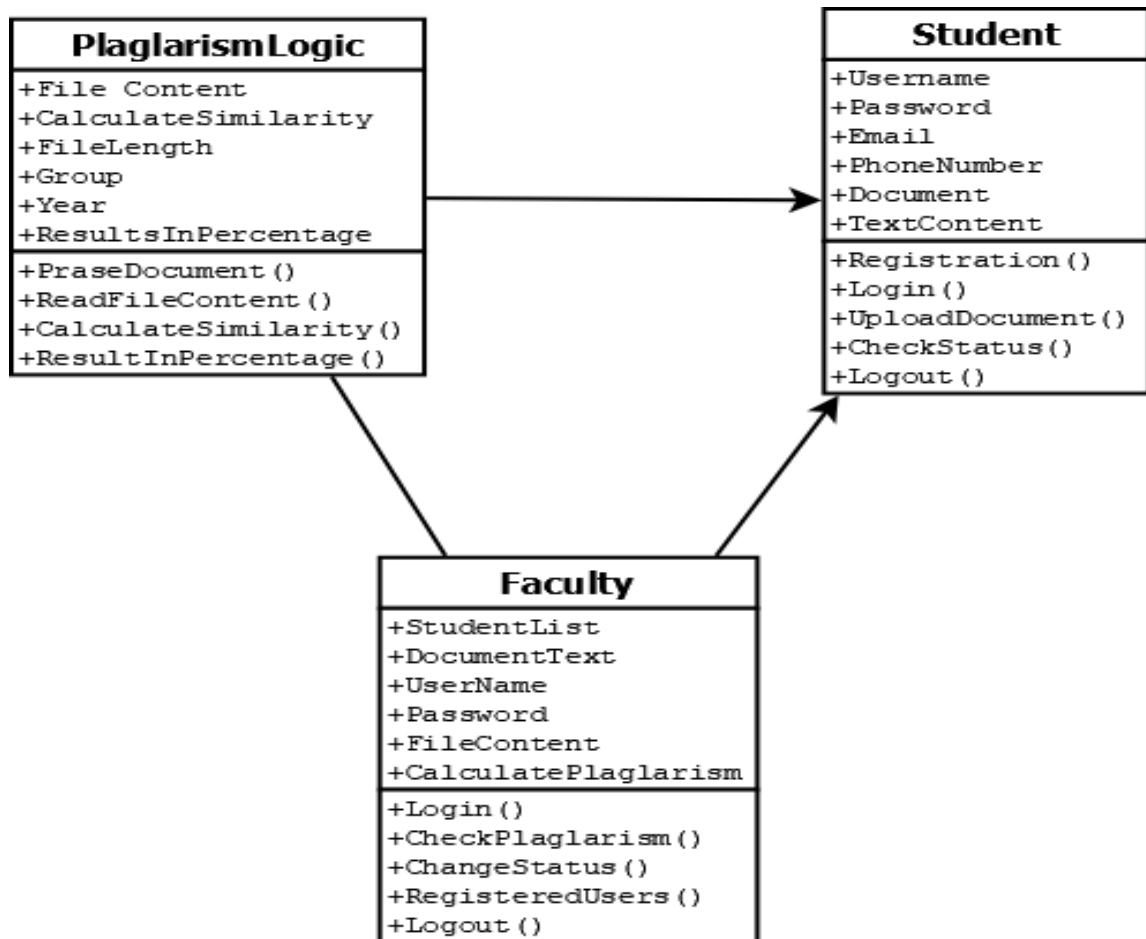


Figure 4.5: Class Diagram

4.6.2 Sequence Diagram

UML Sequence Diagrams are interaction diagrams that detail how operations are carried out. They capture the interaction between objects in the context of collaboration. Sequence Diagrams are time focus and they show the order of the interaction visually by using the vertical axis of the diagram to represent time what messages are sent and when. Sequence Diagram is shown in Figure no.:4.6.

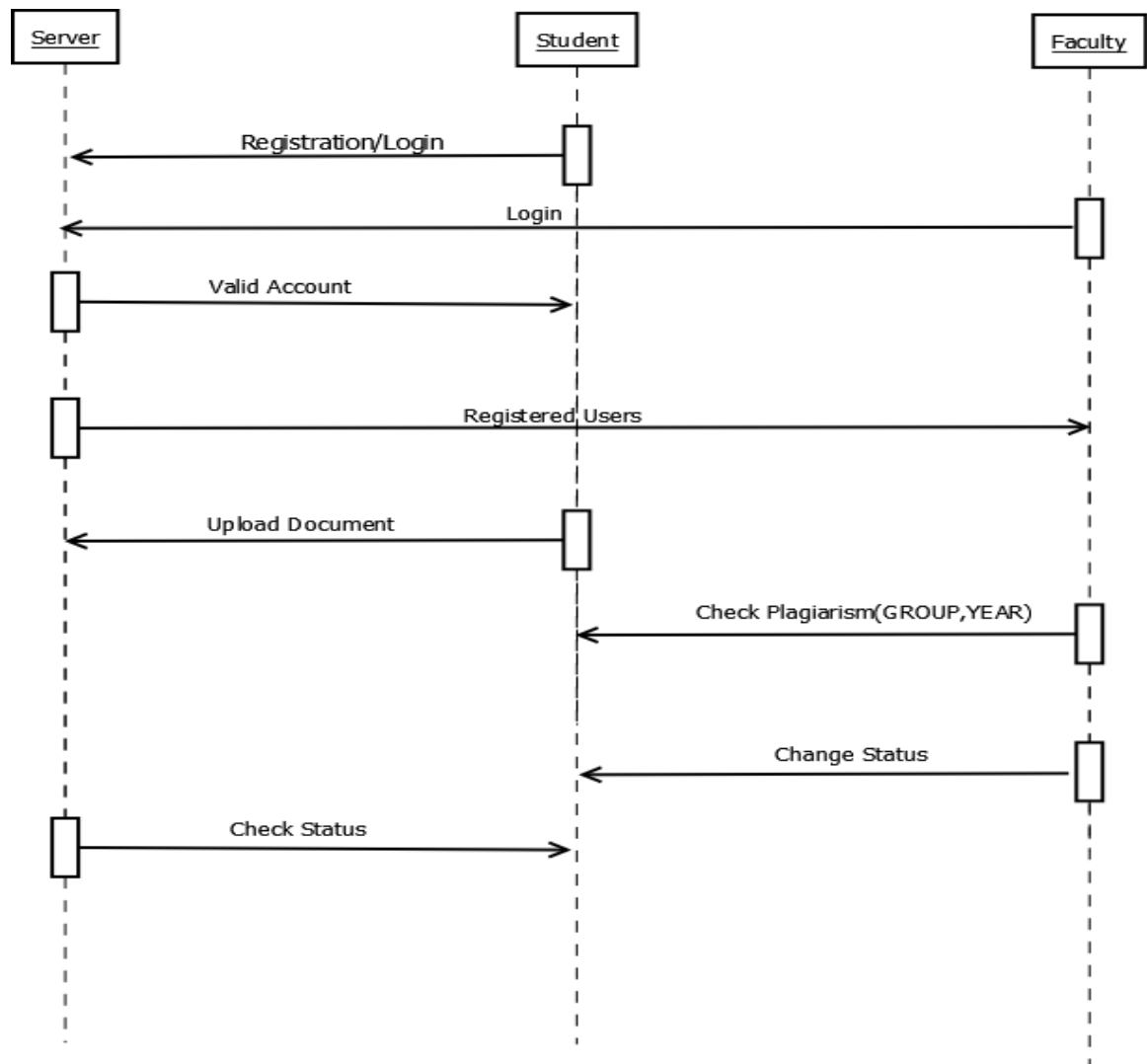


Figure 4.6: Sequence Diagram

4.6.3 Use Case Diagram

A **use case diagram** at its simplest is a representation of a user's interaction with the system that shows the relationship between the user and the different use cases in which the user is involved. A use case diagram can identify the different types of users of a system and the different use cases and will often be accompanied by other types of diagrams as well. The use cases are represented by either circles or ellipses. Use Case Diagram is shown in Figure no.:4.7.

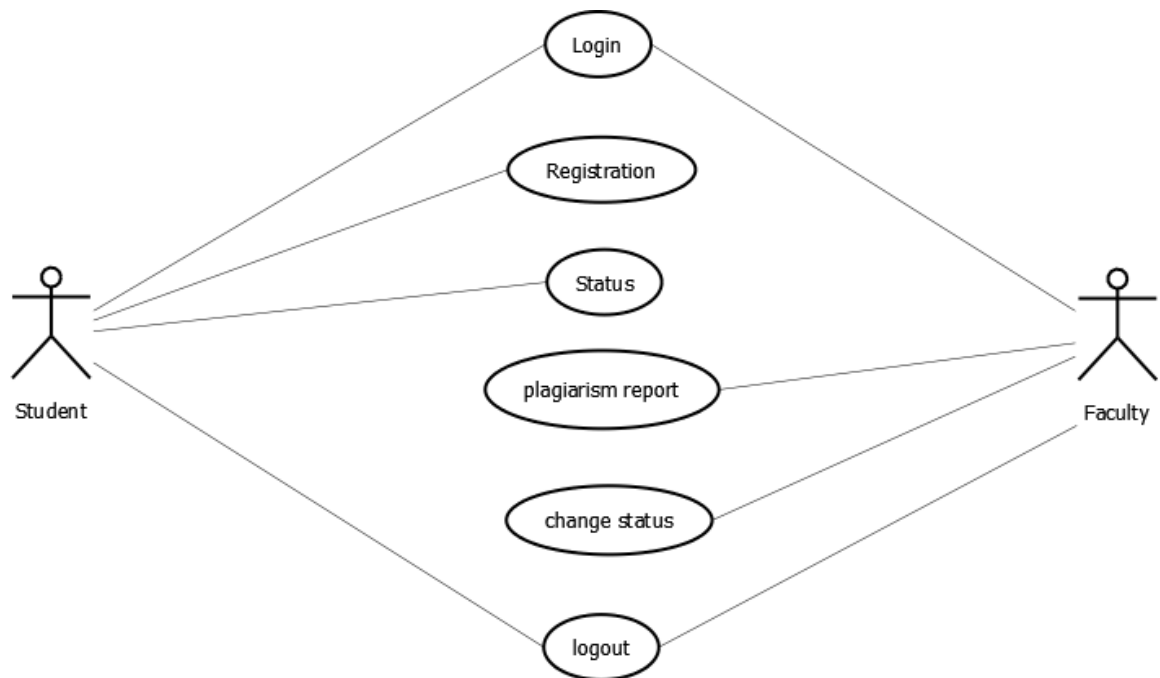


Figure 4.7: Use Case Diagram

4.7 Summary

Design of this application provides clear explanation of all modules and their functionalities with their Dataflow diagrams.

CHAPTER-5

IMPLEMENTATION AND RESULTS

5.1 Introduction

After designing the new system, the whole system is required to be converted into computer understanding language. Coding the new system into computer programming language does this.

It is an important stage where the defined procedures are transformed into control specifications by the help of a computer language. This is also called the programming phase in which the programmer converts the program specifications into computer instructions, which we refer as programs. The programs coordinate the data movements and control the entire process in a system.

It is generally felt that the programs must be modular in nature. This helps in fast development, maintenance and future change, if required.

The validity and proper functionality of all the modules of the developed application is assured during the process of implementation. Implementation is the process of assuring that the information system is operational and then allowing user to take over its operation for use and evaluation.

Implementation is the stage in the project where the theoretical design is turned into a working system. The implementation phase constructs, installs and operated the new system. The most crucial stage in achieving a new successful system is that it works effectively and efficiently.

5.2 System Modules

5.2.1 System Setup Module

Student Module:

Students can registered with valid details like group, Year. Students can able to upload the documents and they can edit the document text. Students can check their plagiarism report along with the status and plagiarism percentage.

Faculty Module:

Faculty can view the system registered students. Check the uploaded documents along with the parsed content. Faculty can able to plagiarized the document and calculate the percentage based on the below following criteria.

1. Based on Group Users
2. Based on Year Users
3. Combination of both Group and Year
4. All Uploaded Documents.

5.2.2 Cross check the document and Parsing Module

To cross check the title of the submitted document files with the database and identify the author of the paper As per IEEE standards no two papers can have the same title unless they have the same author, thus if the title matches then the author of the paper submitted is cross checked with the author of the paper in the database. If the second procedure on inception also gives positive result then the paper is accepted with a new entry with the same title as an update in the author's account. But if the title matches and the author does not match then the paper is rejected.

The further implementation of this module deals with the condition that if the suspected paper is new i.e. in cross checking the title we do not get a positive result. If such is the case then the paper is first checked for plagiarism then after the authenticity of the paper has been ensured the paper is added in the database and when other papers are

cross checked for plagiarism then the paper recently added is also cross referenced with for plagiarism.

5.2.3 Detect Number of words and sentences matched Module

To find the percentage and a logarithmic value to number of words matched in the submitted document. Manipulation of words from a registered paper is one of the plagiarism techniques. Changing the voice (active to passive and vice versa), manipulating the words in the sentences, adding words to the sentences to make it look different than the original one, etc are the techniques used by the culprit to make the paper look like an original work which is not acceptable. In order to remove these anomalies we have built this module.

Text searching technique mentioned above comes handy in this module as well as in the upcoming modules. Let us assume that we have 'X' number of registered documents say $\{D1, D2, D3, \dots, D_x\}$, in our database which are free from plagiarism and each document contains 'n' number of words. Thus total number of words can be given as 'Xn'.

This shows us that we need to sort 'Xn' number of words for every document that will be submitted to the server. By using the Merge sort we can get the least complexity of $(n \cdot \log(n))$ ($xn \cdot \log_2(xn)$) This complexity can be further reduced by selective approach. The registered papers can be divided into finite number of groups and accordingly the submitted paper can be checked against the papers of the group which it belongs to. This module has to return the amount of words matched from a registered paper. These words can either be deliberately added in the submitted document (which can be a case of paraphrasing) or unintentionally. But a certain upper bound must be set in order to maintain the proficiency of the respective topic of research. In this module a logarithmic value and a percentage of the number of words that have been matched are taken into consideration.

This is because the number of words in the submitted document can be huge and taking a direct value cannot be appropriate. Thus percentage to the words copied or a

logarithmic value of the words that are matched comes handy and also it is easy to understand.

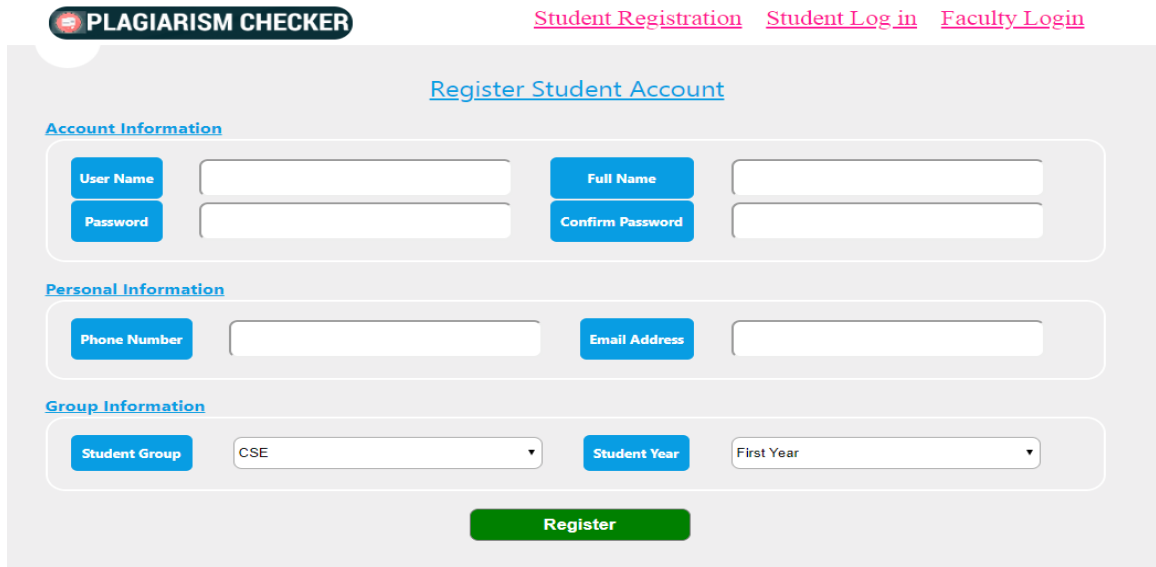
5.2.4 Find the percentage and a logarithmic value Module.

To detect the number of sentences matched in the submitted document. A case of plagiarism can take place when the author completely copies certain number of sentences from the registered document and place these sentences directly to the document which he will submit. To detect this case of plagiarism, we have built this module. This module is mainly dependent on Regular Expressions. A regular expression is a set of pattern matching rules encoded in a string according to certain syntax rules. For every sentence in the submitted document, we need to build a regular expression dynamically in order to match it with the registered documents.

Let us assume that submitted document contains 'n' sentences and every sentence contains 'm' number of words. Thus we need to create 'n' number of regular expressions and these regular expressions will be matched with each of the registered document in the database. The complexity of the algorithm mentioned above will be $O(m)$. Overall there will be 'nm' number of words in the submitted document. Thus to create a set of regular expression for the submitted document will be $O(nm)$. Finally we need to match this set of regular expression with each document in the database.

5.3 Method of Implementation

5.3.1 Output Screens



The screenshot shows the 'PLAGIARISM CHECKER' application interface. At the top, there are navigation links: [Student Registration](#), [Student Log in](#), and [Faculty Login](#). The main heading is 'Register Student Account'. Below this, there are three sections for form input:

- Account Information:** Includes fields for 'User Name', 'Full Name', 'Password', and 'Confirm Password'.
- Personal Information:** Includes fields for 'Phone Number' and 'Email Address'.
- Group Information:** Includes a 'Student Group' dropdown menu (currently showing 'CSE') and a 'Student Year' dropdown menu (currently showing 'First Year').

A green 'Register' button is located at the bottom of the form.

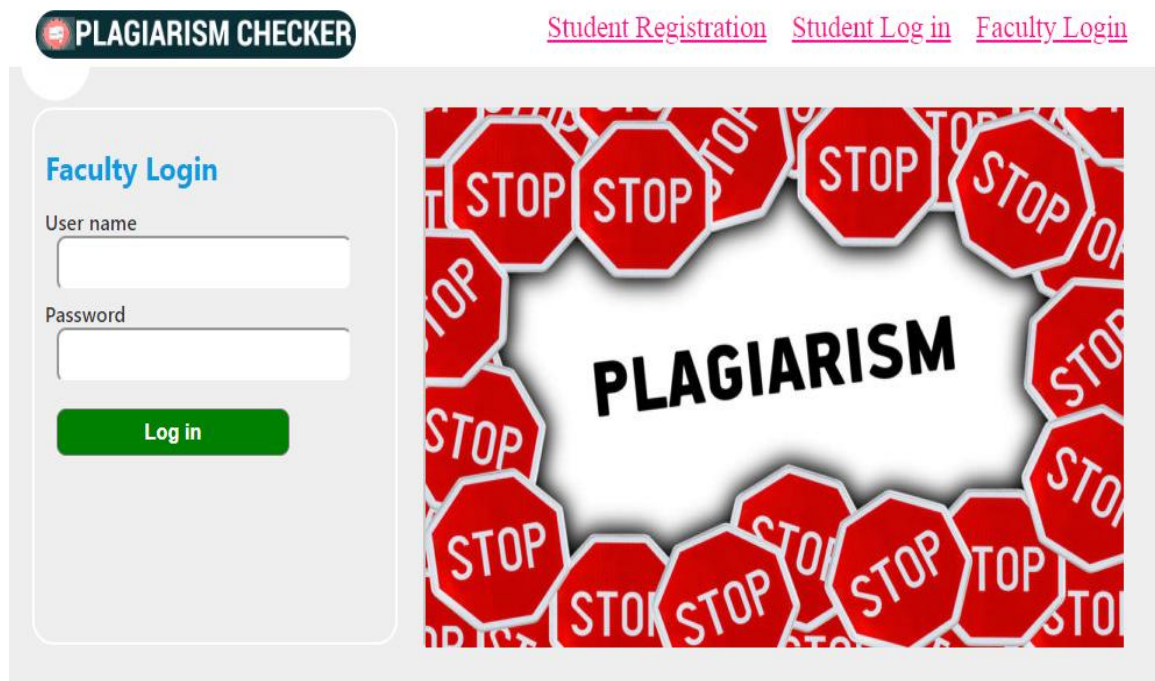
Figure 5.1: Student Registration Page



The screenshot shows the 'PLAGIARISM CHECKER' application interface. At the top, there are navigation links: [Student Registration](#), [Student Log in](#), and [Faculty Login](#). The main heading is 'Student Login'. Below this, there are two input fields: 'User name' and 'Password'. A green 'Log in' button is located below the password field. Below the button, there is a link: [Register](#) if you don't have an account.

The background of the page features a large graphic with the word 'PLAGIARISM' in bold black letters, surrounded by a circular arrangement of red octagonal stop signs.

Figure 5.2: Student Login Page



PLAGIARISM CHECKER

[Student Registration](#) [Student Log in](#) [Faculty Login](#)

Faculty Login

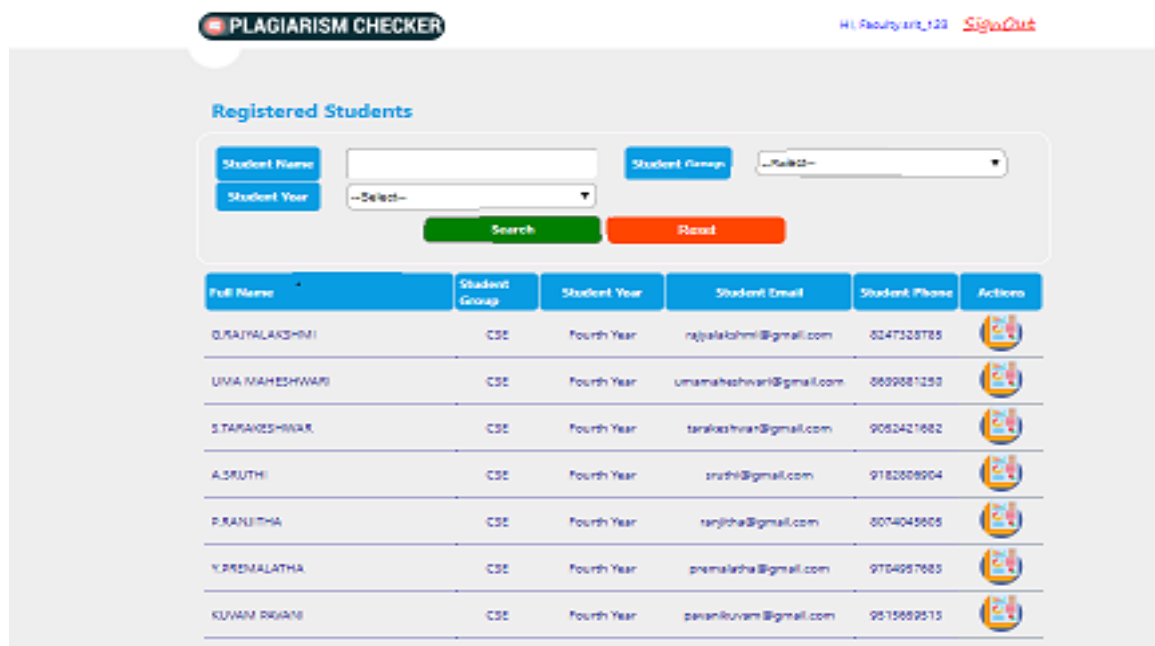
User name

Password

Log in



Figure 5.3: Faculty Login Page



PLAGIARISM CHECKER

Hi, Faculty. [Sign Out](#)

Registered Students

Student Name: Student Group:

Student Year: **Search** **Reset**

Full Name	Student Group	Student Year	Student Email	Student Phone	Actions
QUSAIYALAKSHMI	CSE	Fourth Year	qsaiyalakshmi@gmail.com	9247328785	
UMA MAHESHWARI	CSE	Fourth Year	umamaheshwari@gmail.com	9899581250	
S.TARAKESHVAR	CSE	Fourth Year	tarakeshvar@gmail.com	9052421682	
A.SRUTHI	CSE	Fourth Year	sruthi@gmail.com	9182008904	
P.RANJITHA	CSE	Fourth Year	ranjitha@gmail.com	9074045605	
Y.PREMALATHA	CSE	Fourth Year	premalatha@gmail.com	9704957685	
KUVAM DIVYAN	CSE	Fourth Year	divyanikuvam@gmail.com	9515692515	

Figure 5.4: Faculty Screen

The screenshot shows the 'PLAGIARISM CHECKER' interface. At the top right, it displays 'HI, Faculty.srit_123' and a 'SignOut' link. The main heading is 'Plagiarism Check - KUVAM PAVANI'. Below this, there are two dropdown menus: 'Student Group' set to 'CSE' and 'Student Year' set to 'Fourth Year'. To the right of these is a green 'Check Plagiarism' button. Below the dropdowns, there are two columns: 'Document Text' and 'Percentage Matched'. The 'Document Text' column contains a text area with the following text: 'Cyber Crime: Cybercrime is defined as a crime in which a computer is the object of the crime (hacking, phishing, spamming) or is used as a tool to commit an offense (child pornography, hate crimes). Cybercriminals may use computer technology to access personal information, business trade secrets or use the internet for exploitative or malicious purposes. Criminals can also use computers for communication and document or data storage. Criminals who perform these illegal activities are often referred to as hackers. Cybercrime may also be referred to as computer crime.' Below the text area is a dropdown menu set to '--Change Status--' and a green 'Save Status' button. The 'Percentage Matched' column displays '96.36 %' in large red text.

Figure 5.5: Plagiarism Percentage of the Student

The screenshot shows the 'PLAGIARISM CHECKER' interface. At the top right, it displays 'HI, RAJYALAKSHMI' and a 'SignOut' link. The main heading is 'Upload Plagiarism Document'. Below this, there is a file upload section with a 'Choose File' button and the text 'No file chosen'. Below the file upload section, there are two green buttons: 'Upload Document' and 'Check Status'.

Figure 5.6: Uploading of the Document

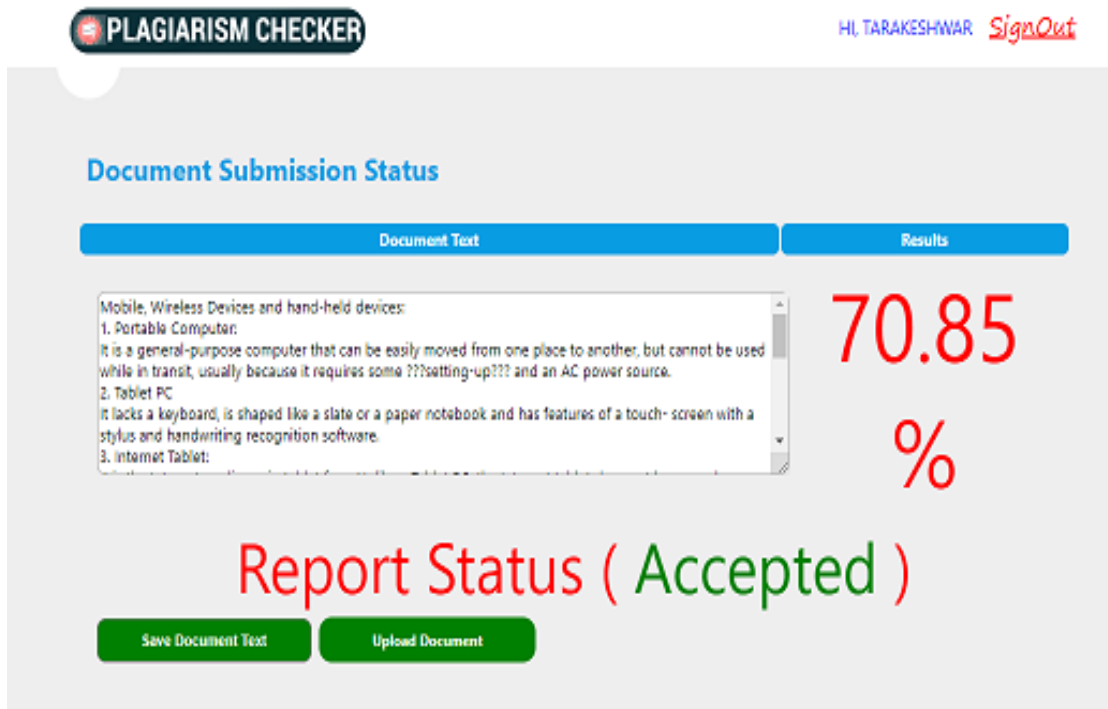


Figure 5.7: Displaying the Status of an Assignment

5.4 Summary

Implementation and results gives the explanation of key functions in Online Assignment Plagiarism Checker application with output screens and dataflow diagrams are used to design this application.

CHAPTER-6

TESTING

6.1 Introduction

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product.

It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

6.2 Types of Tests

Unit Testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration Testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the

combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional Testing

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Testing

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its

purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

6.3 Unit Testing

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases. Test strategy and approach field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

6.4 Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

6.5 Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

6.6 Summary

Testing gives the correctness of the output of the application by using some testing techniques.

CONCLUSION

“Online Assignment Plagiarism Checker” application helps for the faculty, which reduces the manual evaluation of assignments in which it requires much time. To minimize this issue and to help the academic staff in conducting proper evaluation of assignments we introduce this application, which is also helpful for the students to think on their own and it will increase their knowledge as well. It will reduce the time and man power by developing a web application which uses for Plagiarism Checking. In this application the user can register with their basic registration details and create a valid login ID and password and can upload their assignments file. This application compares similarities among all documents and shows the plagiarism percentage, based on the allowable criteria set by the faculty. The faculty can validate whether the document is acceptable or unacceptable.

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- [3] AntiPlag: Plagiarism Detection on Electronic Submissions of Text Based Assignments: <https://scihub.bban.top/https://ieeexplore.ieee.org/document/6732013>